$$\therefore \quad (\vec{b}-\vec{a})\times(\vec{c}-\vec{a}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{vmatrix}$$

$$= \hat{i}+\hat{j}+\hat{k}$$

$$\therefore \quad (\vec{r}-\vec{a}) = (x\hat{i}+y\hat{j}+z\hat{k})-(\hat{i})$$

$$= (x-1)\hat{i}+y\hat{j}+z\hat{k}$$

We have,

$$[(x-1)\hat{i}+y\hat{j}+z\hat{k}].(\hat{i}+\hat{j}+\hat{k}) = 0$$

$$= (x-1)+y+z = 0$$

$\therefore$ Equation of plane $x + y + z - 1 = 0$

Perpendicular distance from origin to the plane

$$= \frac{|-1|}{\sqrt{(1)^2+(1)^2+(1)^2}}$$

$$= \frac{1}{\sqrt{3}} \text{ unit}$$

Here, $p = \dfrac{1}{\sqrt{3}}$

$$\therefore \quad 3p^2 = 3\times\frac{1}{3} = 1$$

57. Option (b) is correct.

We have, $l + 2m + n = 0$ ...(i)

and $2l - 2m + 3n = 0$ ...(ii)

From (i) we have,

$l = -2m - n$ ...(iii)

Put (iii) in (ii)

$$2(-2m-n) - 2m + 3n = 0$$
$$-4m - 2n - 2m + 3n = 0$$
$$-6m + n = 0$$
$$n = 6m$$

Now we have $l = -2m - n$

$$= -2m - 6m = -8m$$

$$\frac{l}{-8m} = \frac{m}{m} = \frac{n}{6m}$$

$$\therefore \quad \frac{l}{-8} = \frac{m}{1} = \frac{n}{6}$$

$$= \sqrt{\frac{l^2+m^2+n^2}{(-8)^2+(1)^2+(6)^2}}$$

$$= \frac{1}{\sqrt{101}}$$

$$\therefore \quad l = \frac{-8}{\sqrt{101}},\ m = \frac{1}{\sqrt{101}},$$

$$n = \frac{6}{\sqrt{101}}$$

$\therefore$ Direction cosines of a line

$$<\frac{-8}{\sqrt{101}},\frac{1}{\sqrt{101}},\frac{6}{\sqrt{101}}> = <l, m, n>$$

$$\therefore \quad l^2+m^2-n^2 = \frac{64}{101}+\frac{1}{101}-\frac{36}{101}$$

$$= \frac{29}{101}$$

58. Option (b) is correct.

Let the equation of any line passing through the point of intersection of the given line be

$$(x + 2y - 1) + a(2x - y - 1) = 0$$

Reducing the equation to its intercept form

$$x + 2y - 1 + 2ax - ay - a = 0$$

$$\Rightarrow \quad (1 + 2a)x + (2 - a)y = 1 + a$$

$$\Rightarrow \quad \frac{(1+2a)}{(1+a)}x+\frac{(2-a)}{(1+a)}y = 1$$

Therefore coordinates of A and B, where this line meets the coordinate axis respectively.

$A = \left(\dfrac{1+a}{1+2a},0\right)$ on x-axis

$B = \left(0,\dfrac{1+a}{2-a}\right)$ on y-axis

Now we need to find the mid-point of AB by using coordinates of A and B

$$\text{Mid-point of } AB = \left(\frac{1+a}{2+4a},\frac{1+a}{4-2a}\right)$$

Now, we find the locus of this point by eliminating 'a' between the two expressions.

Where $x = \dfrac{1+a}{2+4a}$ and $y = \dfrac{1+a}{4-2a}$

$$\Rightarrow \quad 2x + 4ax = 1 + a$$

$$\Rightarrow \quad 4ax - a = 1 - 2x$$

$$\Rightarrow \quad a = \frac{1-2x}{4x-1}$$

We get

$$y = \frac{1+\frac{1-2x}{4x-1}}{4-2\left(\frac{1-2x}{4x-1}\right)}$$

$$y = \frac{4x-1+1-2x}{16x-4-2+4x}$$

$$y = \frac{2x}{20x-6}$$